

```

clear; clc; close all;
addpath(fileparts(mfilename('fullpath')));

params = initialize_paper_parameters();
nRuns = 100;

metrics = struct('positionDeviation', zeros(nRuns,1), ...
    'horizontalVelocityDeviation', zeros(nRuns,1), ...
    'rollDeviation', zeros(nRuns,1), ...
    'pitchDeviation', zeros(nRuns,1), ...
    'yawDeviation', zeros(nRuns,1), ...
    'touchdownTime', zeros(nRuns,1), ...
    'success', false(nRuns,1));

for i = 1:nRuns
    result = simulate_landing_trial(params, i, false);
    metrics.positionDeviation(i) = result.metrics.positionDeviation;
    metrics.horizontalVelocityDeviation(i) = result.metrics.horizontalVelocityDeviation;
    metrics.rollDeviation(i) = rad2deg(result.metrics.rollDeviation);
    metrics.pitchDeviation(i) = rad2deg(result.metrics.pitchDeviation);
    metrics.yawDeviation(i) = rad2deg(result.metrics.yawDeviation);
    metrics.touchdownTime(i) = result.touchdownTime;
    metrics.success(i) = result.success;
    fprintf('Run %3d/%3d: success=%d pos=%.3f m time=%.2f s\n', ...
        i, nRuns, result.success, result.metrics.positionDeviation, result.touchdownTime);
end

save(fullfile(params.paths.results, 'monte_carlo_metrics.mat'), 'metrics', 'params');
plot_monte_carlo_statistics(metrics, params);

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Run    1/100: success=0 pos=56.561 m time=69.90 s
Run    2/100: success=0 pos=57.124 m time=69.90 s
Run    3/100: success=0 pos=53.115 m time=69.90 s
Run    4/100: success=0 pos=42.753 m time=69.90 s
Run    5/100: success=0 pos=54.697 m time=69.90 s
Run    6/100: success=0 pos=43.855 m time=69.90 s
Run    7/100: success=0 pos=57.910 m time=69.90 s
Run    8/100: success=0 pos=52.077 m time=69.90 s
Run    9/100: success=0 pos=60.190 m time=69.90 s
Run   10/100: success=0 pos=52.220 m time=69.90 s
Run   11/100: success=0 pos=57.552 m time=69.90 s
Run   12/100: success=0 pos=58.103 m time=69.90 s
Run   13/100: success=0 pos=50.486 m time=69.90 s
Run   14/100: success=0 pos=52.912 m time=69.90 s
Run   15/100: success=0 pos=47.662 m time=69.90 s
Run   16/100: success=0 pos=57.271 m time=69.90 s
Run   17/100: success=0 pos=55.791 m time=69.90 s
Run   18/100: success=0 pos=52.984 m time=69.90 s
Run   19/100: success=0 pos=59.091 m time=69.90 s
Run   20/100: success=0 pos=50.549 m time=69.90 s
Run   21/100: success=0 pos=64.064 m time=69.90 s
Run   22/100: success=0 pos=59.552 m time=69.90 s
Run   23/100: success=0 pos=53.178 m time=69.90 s
Run   24/100: success=0 pos=48.687 m time=69.90 s
Run   25/100: success=0 pos=47.151 m time=69.90 s
Run   26/100: success=0 pos=57.771 m time=69.90 s
Run   27/100: success=0 pos=54.279 m time=69.90 s
Run   28/100: success=0 pos=52.063 m time=69.90 s
Run   29/100: success=0 pos=47.253 m time=69.90 s
Run   30/100: success=0 pos=50.916 m time=69.90 s
Run   31/100: success=0 pos=54.597 m time=69.90 s

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Run 32/100: success=0 pos=52.474 m time=69.90 s
Run 33/100: success=0 pos=59.151 m time=69.90 s
Run 34/100: success=0 pos=58.365 m time=69.90 s
Run 35/100: success=0 pos=55.623 m time=69.90 s
Run 36/100: success=0 pos=52.575 m time=69.90 s
Run 37/100: success=0 pos=50.889 m time=69.90 s
Run 38/100: success=0 pos=56.220 m time=69.90 s
Run 39/100: success=0 pos=53.544 m time=69.90 s
Run 40/100: success=0 pos=56.719 m time=69.90 s
Run 41/100: success=0 pos=59.975 m time=69.90 s
Run 42/100: success=0 pos=56.607 m time=69.90 s
Run 43/100: success=0 pos=60.201 m time=69.90 s
Run 44/100: success=0 pos=50.340 m time=69.90 s
Run 45/100: success=0 pos=48.789 m time=69.90 s
Run 46/100: success=0 pos=51.127 m time=69.90 s
Run 47/100: success=0 pos=62.130 m time=69.90 s
Run 48/100: success=0 pos=60.647 m time=69.90 s
Run 49/100: success=0 pos=58.262 m time=69.90 s
Run 50/100: success=0 pos=54.875 m time=69.90 s
Run 51/100: success=0 pos=53.689 m time=69.90 s
Run 52/100: success=0 pos=48.368 m time=69.90 s
Run 53/100: success=0 pos=47.998 m time=69.90 s
Run 54/100: success=0 pos=55.260 m time=69.90 s
Run 55/100: success=0 pos=60.598 m time=69.90 s
Run 56/100: success=0 pos=51.548 m time=69.90 s
Run 57/100: success=0 pos=62.069 m time=69.90 s
Run 58/100: success=0 pos=57.471 m time=69.90 s
Run 59/100: success=0 pos=46.720 m time=69.90 s
Run 60/100: success=0 pos=59.306 m time=69.90 s
Run 61/100: success=0 pos=47.288 m time=69.90 s
Run 62/100: success=0 pos=61.531 m time=69.90 s
Run 63/100: success=0 pos=53.460 m time=69.90 s
Run 64/100: success=0 pos=56.440 m time=69.90 s
Run 65/100: success=0 pos=62.012 m time=69.90 s
Run 66/100: success=0 pos=58.084 m time=69.90 s
Run 67/100: success=0 pos=51.935 m time=69.90 s
Run 68/100: success=0 pos=58.888 m time=69.90 s
Run 69/100: success=0 pos=56.045 m time=69.90 s
Run 70/100: success=0 pos=46.041 m time=69.90 s
Run 71/100: success=0 pos=60.383 m time=69.90 s
Run 72/100: success=0 pos=68.675 m time=69.90 s
Run 73/100: success=0 pos=52.207 m time=69.90 s
Run 74/100: success=0 pos=57.094 m time=69.90 s
Run 75/100: success=0 pos=54.302 m time=69.90 s
Run 76/100: success=0 pos=56.625 m time=69.90 s
Run 77/100: success=0 pos=51.618 m time=69.90 s
Run 78/100: success=0 pos=64.407 m time=69.90 s
Run 79/100: success=0 pos=54.037 m time=69.90 s
Run 80/100: success=0 pos=53.129 m time=69.90 s
Run 81/100: success=0 pos=56.660 m time=69.90 s
Run 82/100: success=0 pos=55.319 m time=69.90 s
Run 83/100: success=0 pos=57.379 m time=69.90 s
Run 84/100: success=0 pos=61.581 m time=69.90 s
Run 85/100: success=0 pos=51.139 m time=69.90 s
Run 86/100: success=0 pos=65.785 m time=69.90 s
Run 87/100: success=0 pos=49.523 m time=69.90 s
Run 88/100: success=0 pos=51.665 m time=69.90 s
Run 89/100: success=0 pos=54.959 m time=69.90 s
Run 90/100: success=0 pos=62.117 m time=69.90 s
Run 91/100: success=0 pos=64.625 m time=69.90 s
Run 92/100: success=0 pos=44.225 m time=69.90 s
Run 93/100: success=0 pos=52.035 m time=69.90 s
Run 94/100: success=0 pos=49.562 m time=69.90 s
Run 95/100: success=0 pos=58.182 m time=69.90 s

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Run 96/100: success=0 pos=57.399 m time=69.90 s
Run 97/100: success=0 pos=45.960 m time=69.90 s
Run 98/100: success=0 pos=51.655 m time=69.90 s
Run 99/100: success=0 pos=51.262 m time=69.90 s
Run 100/100: success=0 pos=54.230 m time=69.90 s

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Monte Carlo summary:

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success rate: 0.0 %
position deviation N(54.834, 5.122^2)
horizontal velocity N(0.291, 0.144^2)
roll deviation N(0.228, 1.884^2) deg
pitch deviation N(-0.812, 3.436^2) deg
yaw deviation N(-19.677, 0.220^2) deg

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